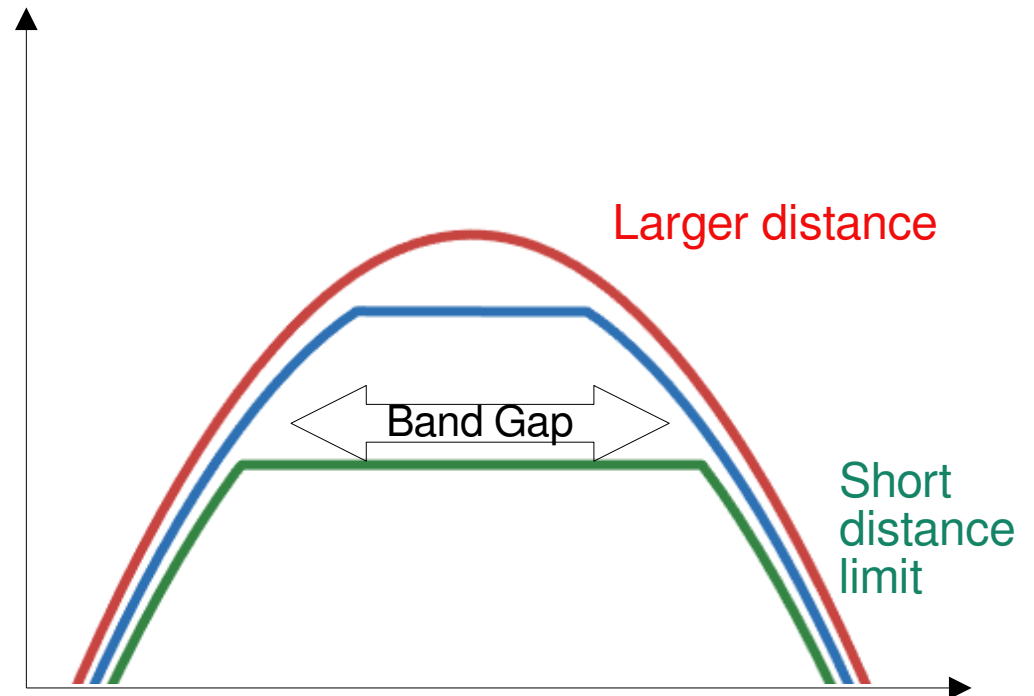
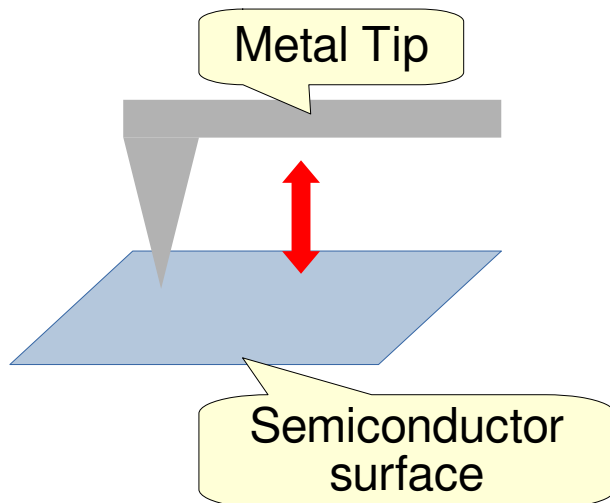


Felix Mo

- Modelling the capacitance across a Metal-Insulator-Semiconductor device
- Application: Future experiment on determining the local band gap of semiconductor materials
- Idea for this project proposed by John Marohn
- Python package: <https://github.com/fm466/semiconductor-tip-capacitor-model-fm466.git>

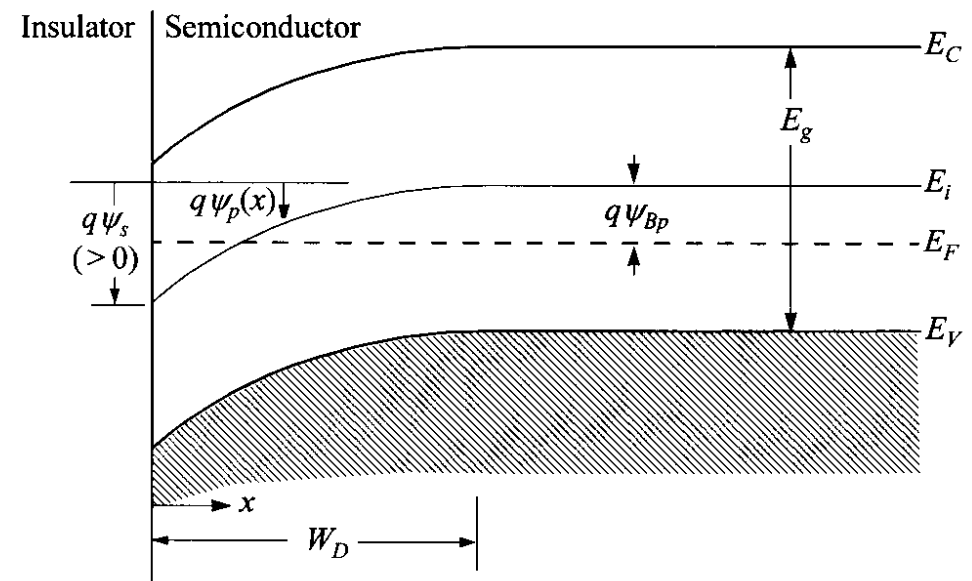
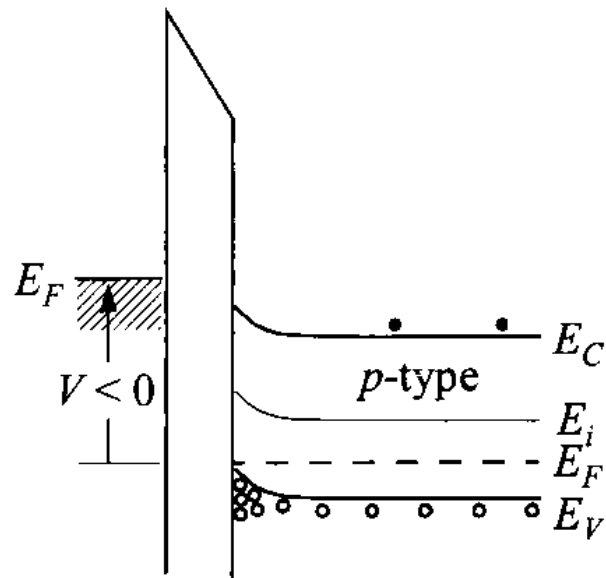
Background

- Model an experiment to determine local band gap of semiconductor materials across small areas
- Based on the principle of metal-insulator-semiconductor devices



Theory

- In the surface space-charge region, band bending takes place as voltage is applied to the system
- Modelling this band bending and resultant surface potential by solving Poisson equation



Model

- Capacitance is modelled, which would be measured indirectly by change in cantilever spring constant or frequency shift during an experiment
- Accepts variables:
 - Voltage range
 - Temperature
 - Semiconductor band gap and electron affinity
 - Insulator / Semiconductor permittivity
 - Conduction / Valence band density of states
 - Donor / acceptor density
 - Metal work function

